



AI & CYBERSECURITY HACKATHON | IBM WATSONX TRACK

CyberSentinel — AI-Powered SOC Analyst

Autonomous Threat Detection · Correlation · Explainable Risk Scoring · Incident Response

Team Xsquade · Lead: **Jenil Sakhiya** · Daksh Kachhadiya · Yug Kansagara · Dhruv Bakhlakiya

Made with GAMMA

The Problem: Alert Overload Is Breaking SOCs

Enterprise SOCs and Tier-1/Tier-2 analysts face **10,000+ fragmented daily alerts** across firewalls, identity providers, and endpoints — with no unified triage.

Alert Fatigue

Analysts burn out chasing disconnected false positives across siloed tools.

Investigation Bottleneck

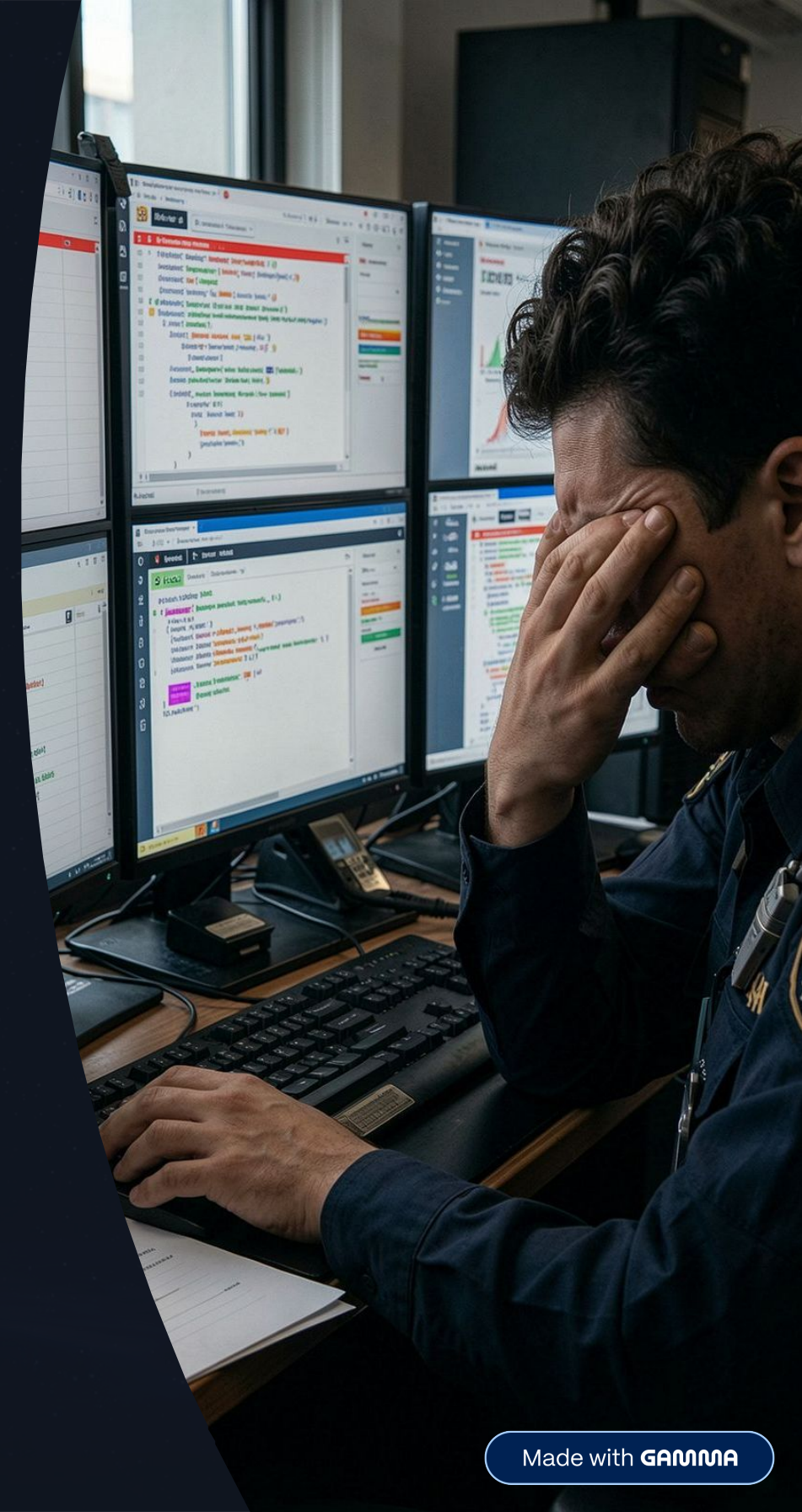
30–45 minutes of manual triage per incident across disjointed consoles.

Attacker Dwell Time

Delayed containment enables lateral movement, privilege escalation, and data theft.

Black-Box Opacity

Traditional tools assign arbitrary risk scores without verifiable evidence.



The Solution: Autonomous AI SOC Analyst

CyberSentinel replaces manual log stitching with a **5-stage autonomous pipeline** — from raw telemetry to explainable incident response.



Telemetry Ingestion

Auth logs, endpoint execution, privilege changes, egress flows



Detection Engine

6 transparent rules: brute-force, unrecognized device, priv-esc, data exfil



Multi-Vector Correlation

Groups events by user, IP, device within 30-min windows



Explainable Scoring

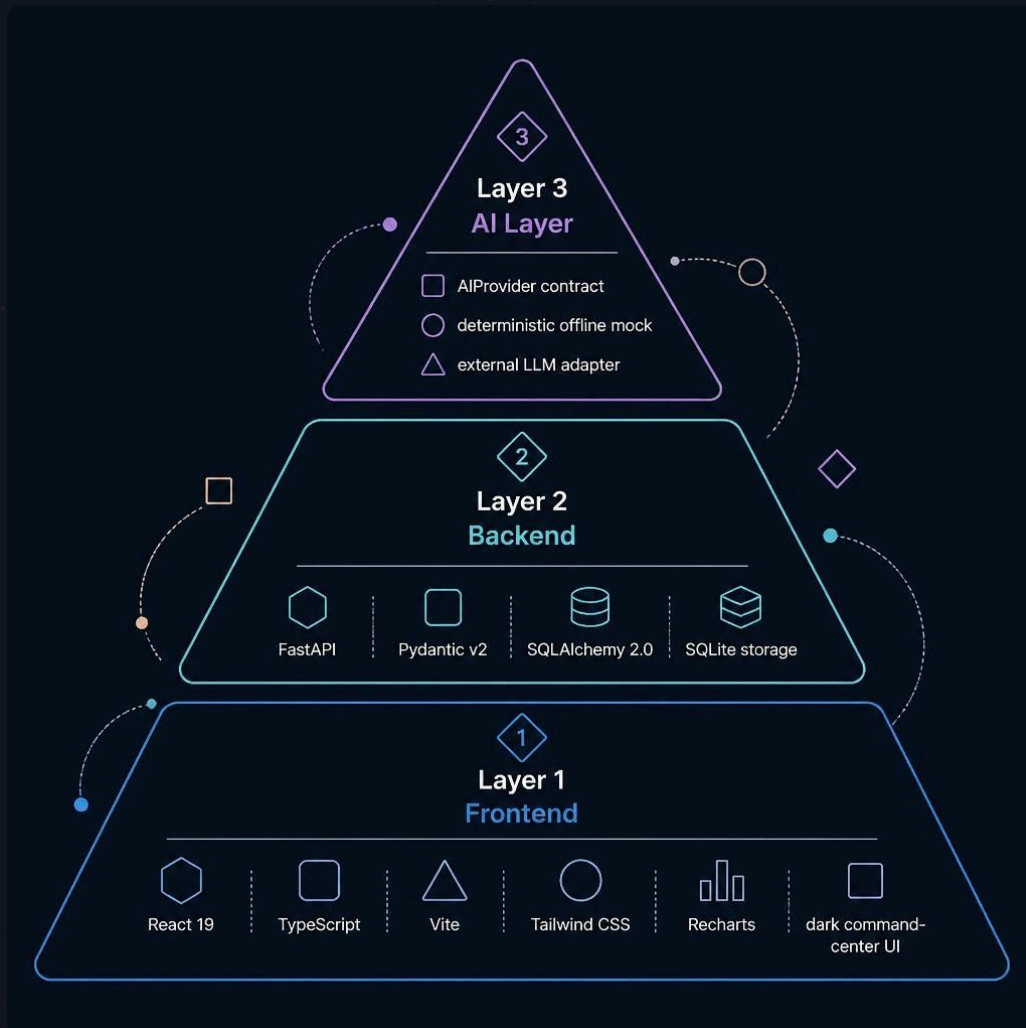
0–100 scale with itemized, transparent point weights



AI Investigation

MITRE ATT&CK hypotheses + grounded interactive Q&A

Architecture & Demo Highlights



Full Attack Chain — 1-Click Simulation

Trigger a complete adversarial scenario in a single click:

Real-time clustering auto-classifies the chain as **CRITICAL (Score 90–100)** with TLP:AMBER Markdown/JSON report exports.

01

Brute Force Attack

02

Compromised Login

03

Privilege Escalation

04

1.8 GB Data Exfiltration

IBM watsonx & Bob Integration

IBM Granite (granite-3.0-8b-instruct)

Powers the deep investigation hypothesis engine — synthesizing attack patterns and generating plain-English danger assessments grounded in incident telemetry.

IBM Granite Guardian

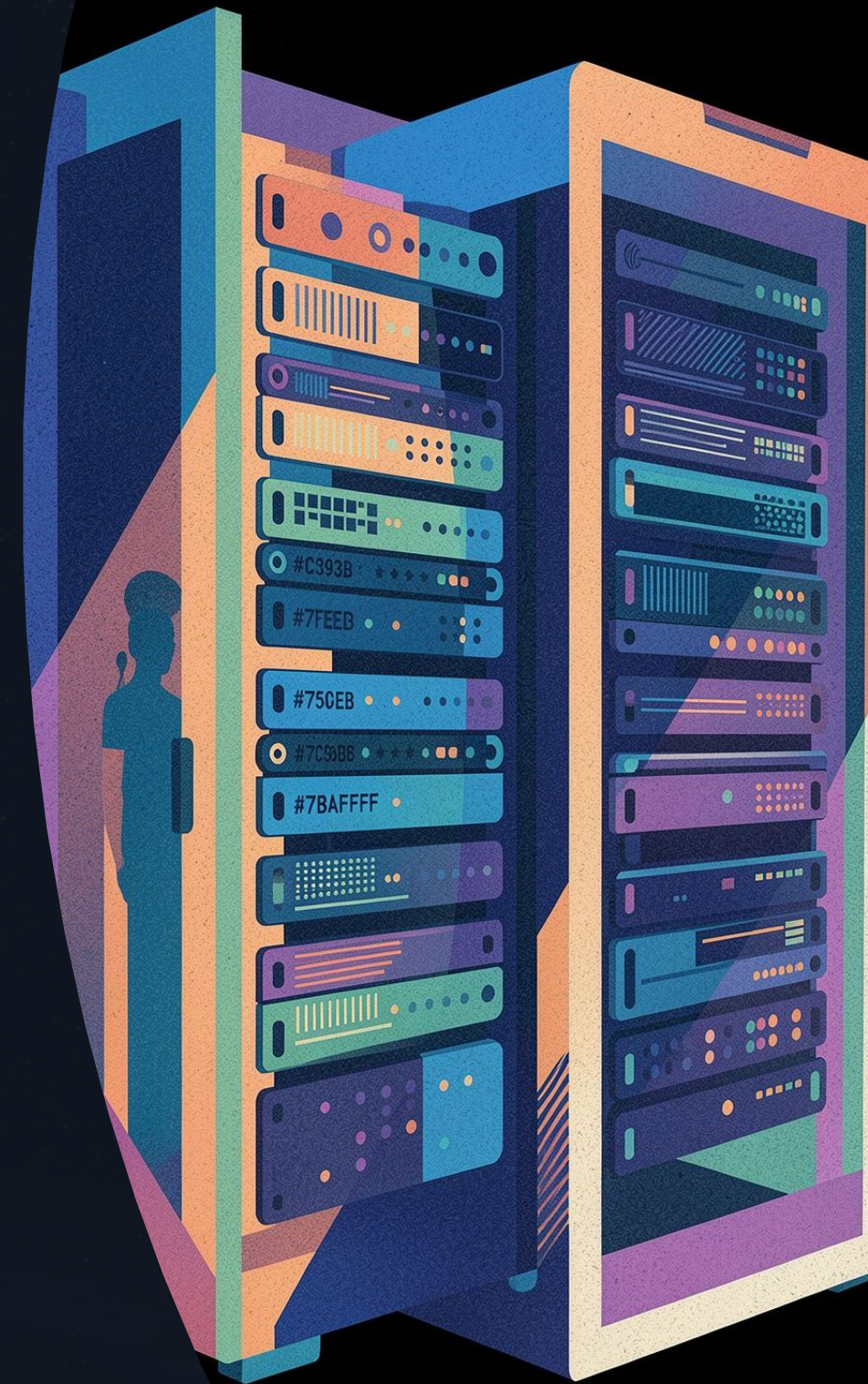
Provides safety filtering and **zero-hallucination guardrails**, ensuring every AI response strictly cites verified incident evidence.

IBM Bob — Code Assistant

Accelerated architecture scaffolding, deterministic rule definition, and temporal correlation logic. Automated 16 unit/integration tests — **100% pass rate in under 1.5 seconds.**

IBM QRadar SIEM / SOAR Alignment

Designed as an intelligent triage agent directly compatible with IBM QRadar telemetry pipelines for seamless enterprise deployment.



Impact & Future Roadmap

85%

MTTR Reduction

Incident triage drops from 45 minutes to under 5 minutes

90%

Alert Fatigue Cut

Intelligent clustering condenses hundreds of raw alerts into single actionable incidents

100%

Explainable Auditing

Zero black-box decisions — every score is transparent, itemized, and compliance-ready

What's Next



Automated SOAR Orchestration — Safe containment via Okta session revocation and CrowdStrike host isolation



Enterprise Threat Intel — STIX/TAXII live feeds + fine-tuned watsonx models on proprietary security data



Multi-Tenant Cloud Scale — TimescaleDB/PostgreSQL with distributed streaming via Kafka/Redpanda

Project Summary

CyberSentinel is a **fully functional, end-to-end autonomous SOC analyst** — built for real enterprise security operations, not just a demo.

Complete Implementation

Frontend, Backend, AI engine, Attack Simulator, and Reporting — all shipped

Rigorous Engineering

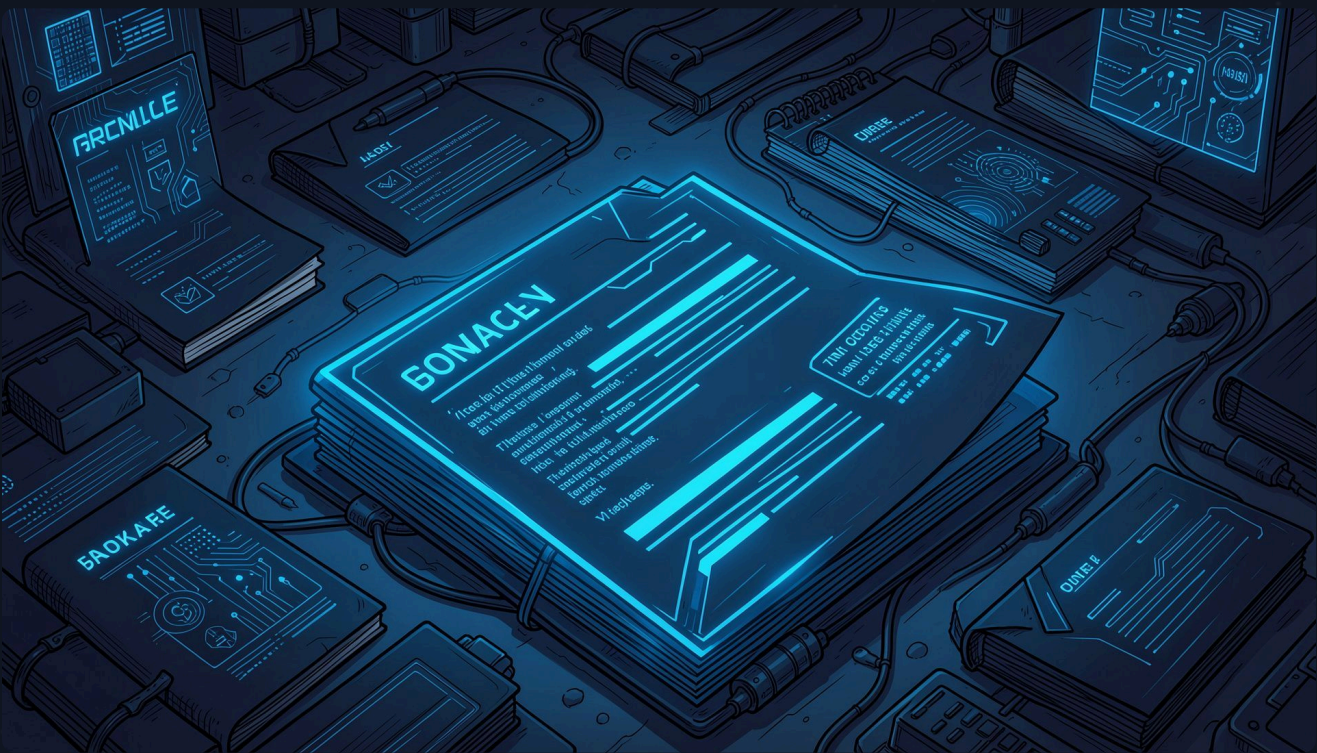
16 passing tests, clean decoupled code, full offline portability

Live Demo Ready

1-click attack chain with real-time clustering and explainable AI reasoning



Presentation Deliverables



Generated Artifacts

	presentation/slides.m d Marp markdown deck with custom dark cyber CSS theme
	presentation/generat e_pptx.py Automated python-pptx generator with 16:9 widescreen layout
	presentation/slides.p df High-definition PDF compiled via Marp CLI
	presentation/slides.p ptx Full PowerPoint deck for Microsoft PowerPoint / Google Slides

Verification Plan

Automated Verification

1 Run generate_pptx.py

Execute with Python 3.10 to build
presentation/slides.pptx

2 Compile PDF via Marp

Run `marp --pdf --allow-local-files` to produce
presentation/slides.pdf

3 Validate Output

Verify file generation, size, slide count, and formatting

Visual Verification Checklist

Problem Slide

Alert fatigue, bottleneck,
dwell time, opacity
clearly displayed

Solution Pipeline

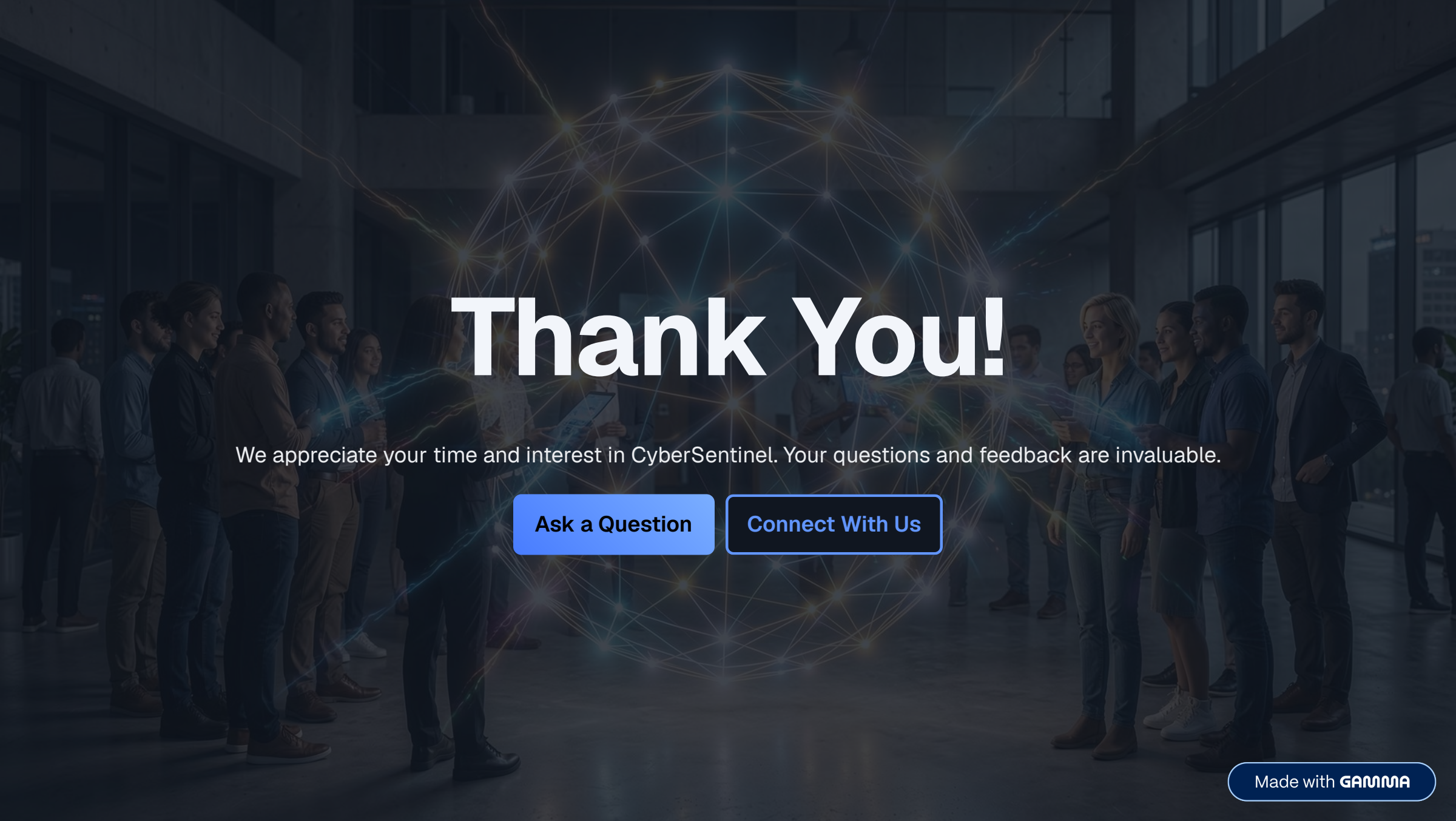
5-stage flow from
ingestion to AI
investigation

IBM Integration

Granite, Guardian, Bob,
and QRadar alignment
featured

Impact Metrics

85% MTTR, 90% alert
fatigue, 100%
explainability

A group of people in a modern office setting, overlaid with a glowing network diagram. The background is a dark, modern office interior with large windows and people standing in a line. A glowing network diagram, consisting of interconnected nodes and lines, is superimposed over the scene, centered around the text.

Thank You!

We appreciate your time and interest in CyberSentinel. Your questions and feedback are invaluable.

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